

Mohammed Zuhair Hussain

Artificial Intelligence Engineer

mohammedzuhairhussain28@gmail.com | +91 9701007757 | github.com/zssain | linkedin.com/in/zuhairhussain28

Education

B.Tech in Artificial Intelligence Engineering Mahindra University CGPA: 8.2/10.0	Sept 2022 – 2026 (Expected)
Intermediate Education FIITJEE Junior College Achieved 94.4% in XII Board Examination	Apr 2020 – Mar 2022
Secondary Education FIITJEE World School Secured 10 CGPA in X State Board Examination	Jun 2017 – Mar 2020

Relevant Coursework

Courses: Data Structures & Algorithms, Design & Analysis of Algorithms, Optimisation Techniques for AI, Machine Learning, Database Management Systems, Computer Networks, Operating Systems, Digital Image Processing, Deep Neural Networks, Numerical Methods, Probability & Statistics, Discrete Math, Foundation of NLP

Work Experience

AI/ML Engineer Intern — Teclusion AI	<i>September 2025 – Present</i>
- Built a Retrieval-Augmented Generation (RAG) chatbot over enterprise documents that retrieves and grounds answers in organizational knowledge sources, automating internal information retrieval and improving knowledge accessibility. - Engineered the underlying Intelligent Document Processing (IDP) system, integrating SharePoint, Google Drive, and enterprise data lakes for centralized document ingestion, indexing, and contextual search across repositories — the retrieval layer powering the chatbot. - Built a dynamic pricing engine using machine learning that analyzed internal sales signals and competitor-scraped pricing data to recommend optimal product prices, with web-scraping pipelines to collect and process competitor data.	
AI Intern, Emerging Tech — PricewaterhouseCoopers	<i>June 2025 – August 2025</i>
- Developed a full-stack AI-driven financial advisory platform (FastAPI, Next.js, Firebase) with an Agentic AI architecture using Google Gemini and a custom Knowledge Graph, enabling real-time, personalized investment guidance, tax optimization, and contextual financial reasoning, with answers grounded against the Knowledge Graph and a median response time under 2s. - Co-led the authoring of a white paper on AI-enabled software development, driving core research and framework design on LangChain, vibe-coding, and agentic AI; collaborated closely with a co-author to present SDLC optimization using generative AI.	

Projects

Agentic AI-Powered Recruitment Automation Platform Built an agentic, multi-agent system that orchestrates specialized AI agents to automate CV scoring, job-description optimization, bias detection, and explainable feedback for candidate selection Designed and implemented the agent orchestration backend in Python, with modular agents for fairness assessment, performance grading, and explainability Built an interactive Streamlit dashboard to surface candidate evaluations, fairness scores, and AI-driven recommendations in real time Integrated CSV-based dynamic data pipelines and adjustable feedback loops for iterative improvement of agent decisions Technologies: Python, Agentic AI (multi-agent orchestration), Explainable AI, Streamlit	GitHub
Multi-Model Image Colorization System with Interactive Dashboard Engineered a sophisticated image colorization system using custom-built models with OpenCV and Autoencoder, leading to an impressive PSNR of 30.02 dB and SSIM of 0.97 in performance evaluations	GitHub

- Designed a dashboard using **Streamlit** and **Gradio** for real-time image uploads and performance evaluation
- Implemented **image preprocessing and post-processing pipelines** to improve model stability and enable consistent colorization across diverse grayscale image datasets.

Technologies: Python, PyTorch, TensorFlow, OpenCV, Streamlit, Gradio

• WISP Offline Mesh Messaging PWA

GitHub

- Architected a **serverless Progressive Web App** for crisis communication across **three transport tiers** (WebRTC over Wi-Fi, animated QR sync, printed posters), deployable via URL in **under 30 seconds** with zero backend
- Engineered a **gossip-style sync engine** with **Bloom filter anti-entropy**, **Ed25519 signatures**, and **NaCl end-to-end encryption**, achieving **sub-200ms LAN sync** with chain-of-custody hop stamping and panic-wipe identity rotation
- Designed an **offline-first** architecture with IndexedDB persistence, TTL-based auto-cleanup, and a live force-directed mesh visualization in a **180KB gzipped bundle**

Technologies: React 18, TypeScript, WebRTC, IndexedDB (Dexie), Ed25519, TweetNaCl, Vite, Service Workers

Skills

Programming Languages: Python, TypeScript, Java, C, SQL

Areas of Expertise: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLM Applications

Deep Learning Frameworks: PyTorch, TensorFlow, Keras

LLM Tooling: Hugging Face Transformers, LangChain, LangGraph

Databases: PostgreSQL, NeonDB, Supabase, MongoDB, Firebase, MySQL

Web Development: FastAPI, Node.js, Express.js, React.js, Next.js, React Native, HTML, CSS, Streamlit, Gradio

Data & Automation: Web Scraping (Selenium, BeautifulSoup), Data Pipelines, API Development, REST APIs

Cloud & Deployment: AWS, Azure, Vercel

Version Control & Environments: Git, GitHub, Google Colab

Libraries & Tools: NumPy, Pandas, Scikit-learn, NLTK, SpaCy, Matplotlib, Seaborn, OpenCV

Awards & Achievements

- **Finalist** — Synchrony UniHack 2025
- Awarded an academic **Scholarship** worth \$1,250 (INR 100,000) for excellent academic performance

Positions of Responsibility

General Secretary — Mahindra University

June 2024 – June 2025

- Acted as the key liaison between students and university management, **addressing operational issues, resolving concerns, and promoting collaborative problem-solving during events**
- Increased the number of club activities across the university, driving **higher student participation** and building a **vibrant campus culture**
- **Systemized the administrative structure, streamlining workflows** and **enhancing the efficiency** of event management and communication processes
- **Led and organized 5 major campus events** including **Farewell, Orientation, Freshers, Convocation** (for **1,000+ graduates**), and the cultural event "**RRR**," fostering student engagement and campus spirit
- **Collaborated** with organizing committees during initial planning and groundwork of **AEON**, the university's first-ever Tech Fest, contributing to the **conceptualization of events, the budgeting, and the formation of the core team**
- **Managed a team of 16 members**, overseeing **event planning, task delegation, logistics coordination, and on-ground execution** to ensure successful event delivery